Continuously Variable Multi-Permeability Inductor for Improving the Efficiency of High-Frequency DC–DC Converter

Junchang Liu, Yunhui Mei, *Member, IEEE*, Shengchang Lu, Xin Li, and Guo-Quan Lu, *Fellow, IEEE*

***Abstract*—This paper proposed a method for one step fabricating the continuously variable multi-permeability (CVMP) toroid core by applying a specific gradient sintering temperature from the inner wall to the outer wall, i.e., along the radius direction, of a toroid core in a tube furnace using a high-temperature resistant stainless steel pipe which is used for flowing the cooling medium. The sintering temperature gradient was designed based on the relationship between permeability and sintering temperature. A CVMP core with the permeability gradually increasing continuously along the radius could be made in this way. It is beneficial for improving the uniform distribution of magnetic flux density of the core by this way. The distribution of magnetic flux density and the inductance was analyzed in theoretical calculation and finite-element analysis. A CVMP core and a single-permeability core were fabricated to evaluate their performance using an 8-V input, 1-V output dc–dc converter. The results show that the CVMP core is possible to balance the inductance and improve the output efficiency compared with a single-permeability core employed in a dc–dc converter at heavy load. It can be concluded that the CVMP core has the capacity of taking higher current than a single-permeability core.**

***Index Terms*—Constant flux, core loss density, efficiency, multi-permeability, toroid core.**

I. Introduction

NOWADAYS, with the development of power electronic technology, the power conversion devices, such as dc–dc converters or transformers, play an important role in the course of energy conversion and transmission [1]–[5]. The inductor, as the crucial component of dc–dc converter, has attracted a lot of attention in recent years due to the fact that it is important to increase the power density and reduce power loss of the dc–dc converter [6]–[9]. Therefore, there is a new demand for the inductor with high inductance, low loss density to increase the power density since the inductor usually occupies a significant percent of the volume and the weight in a converter [10]–[15].

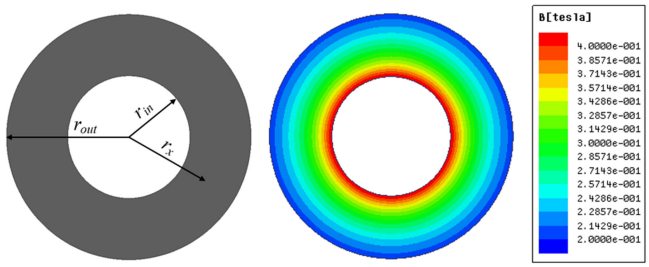


Fig. 1. Schematic diagram and the distribution of magnetic flux density of a commercial toroid core.

The commercial toroid cores are preferred for making magnetic materials with high permeability to obtain high inductance [16]. It should also be noted that it is inappropriate to use high permeability materials for high operation frequency applications due to premature magnetic flux saturation. Therefore, it is vital to select a suitable magnetic materials with appropriate permeability for a specific operation frequency [17]. The permeability of a commercial toroid core is usually identical since the core is composed of the same magnetic materials. Therefore, the magnetic flux density (B) is not uniform from the inner side to the outer side of the core. It varies as a function of the radius according to (1) [17]. The schematic diagram and the distribution of magnetic flux density of the commercial toroid core are shown in Fig. 1

$$B = \mu_0 \mu_r H = \frac{\mu_0 \mu_r I}{2\pi r} \tag{1}$$

where μ_0 is the permeability of vacuum, μ_r is the relative permeability of the core, I is the exciting current flowing through the winding, and r is the radius of the core. The magnetic flux density (B) is inversely proportional to the radius, which causes the non-uniform distribution of magnetic flux in the core. Consequently, the inner wall of the toroid core will be saturated prematurely rather than the outer wall, which is not beneficial to increase the inductance at the larger current. The saturation of magnetic cores could also result in massive loss heat, which

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contributes to the lower output efficiency of the converter. Moreover, the power loss becomes more serious and obvious in high frequency. Therefore, it is necessary to design and fabricate the cores which can take higher current and higher frequency simultaneously.

In order to avoid the premature saturation of the inner wall of the toroid core, it should be interesting if we could design and make a core with variable permeability versus the radius to realize uniform magnetic flux density distribution as well as reducing the loss density. Many studies had been done to target for uniform magnetic flux density distribution. There were usually two strategies considering B is related to μ_r and H, as shown in (1). First, it is possible to obtain the uniform magnetic flux density distribution by adjusting the μ_r in a core. People [18]–[20] had used two or three kinds of materials with different permeability to fabricate two-cell or three-cell, named multi-permeability inductor instead of the commercial inductor with only one magnetic material. In this way, the materials with low permeability and high permeability were used to the inner side and outer side of the inductor, respectively. The inductance and the output efficiency employed in a dc–dc converter of the multi-permeability inductor were both increased than that of a single-permeability inductor [19]. However, the permeability of each cell is discrete, leading to the variable B between the different cells. In addition, the different shrinkage among the cells, which was caused by the thermal expansion coefficient mismatch between different materials, could easily lead to the occurrence of crack or distortion during the cofiring process.

In the aspect of H, others [21]–[24] had developed "constant-flux" inductors to achieve uniform magnetic flux density by designing the windings and radius of each core cell. The permeability of each core cell is identical since the same magnetic materials is used. As demonstrated in [21], the inductor consists of a spiral winding embedded in a magnetic core, and the winding width and spacing have to be precisely varied for each turn of the spiral. In [24], H was adjusted by changing the turns of windings wound in each core cell. In conclusion, the relative uniform flux distribution could be obtained by changing the external magnetic field (H) versus the radius of the core, as shown in (1). H near the core center was set to be lower than that far away from the core center. In this way, the uniform flux distribution could be achieved, but it takes more steps to wind the coppers to each core cell than a conventional toroid core and leads to a complicated process. In a word, there has not been a good way to solve the problem by taking both the properties and complex process into account. And there is no good way to realize uniform magnetic flux density distribution of a toroid core simply.

In this paper, we proposed an idea for one step fabricating a continuously variable multi-permeability (CVMP) toroid core to realize uniform magnetic flux density distribution using only one magnetic material by gradient sintering, rather than the discrete multi-permeability inductors using many different magnetic materials compared with [18] and [19]. The sintering temperature was defined as the temperature to fuse the ferrite particles *in situ* for generating a solid toroid core. The sintering temperature of outer wall of the CVMP core is the setting temperature of the tube furnace. The sintering temperature of the inner wall could be adjusted by controlling the flowing rate of the coolant. The CVMP not only can solve the problem of the discrete permeability caused by different materials but also simplify the preparation process compared with [19] and [21]. The potential defects, e.g., cracks, among the discrete core cells due to the mismatch of the cell shrinkage could be almost eliminated compared with [18]. The crowding of the winding wound in the core cells could also be avoided compared with the "constant-flux" inductors mentioned by Xu and Ngo [24].

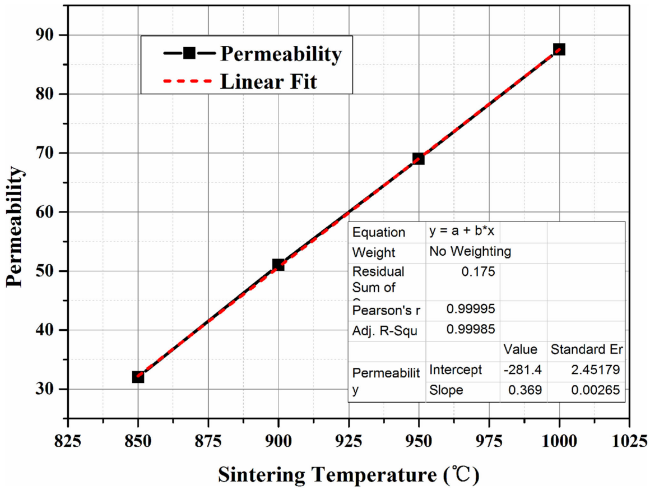


Fig. 2. Permeability of a NiZnCu ferrite as a function of sintering temperature.

In this paper, Section II describes the principle and how to design a CVMP toroid core with rectangular cross section used only one magnetic material. Section III simulates the flux density distribution, core loss density distribution as well as the inductance versus dc of a single-permeability core and a multi-permeability toroid core by finite-element analysis. Section IV describes the processing method to fabricate the CVMP toroid core based on our design as a demonstration. The performance of the demonstrated CVMP core and the single-permeability core is also characterized separately when they are used in a dc–dc converter. Section V gives the conclusion.

II. Principle of CVMP Toroid Cores

This section presents the design principle of the CVMP toroid cores with rectangular cross section. It was found that the permeability almost increased linearly with the sintering temperature from 850 to 1000 °C [25]. Therefore, it is possible to design a multi-permeability toroid core using only one magnetic material by sintering the core under a gradient temperature profile. It is reasonable to obtain a CVMP toroid core with uniform magnetic flux density by controlling the sintering temperature along the radius direction of the toroid core precisely. For example, a numerical relationship between the permeability and the sintering temperature of a typical NiZnCu ferrite is present as (2) and Fig. 2 [25]

$$\mu_r = 0.369T - 281.45 \qquad 850 \le T \le 1000. \tag{2}$$

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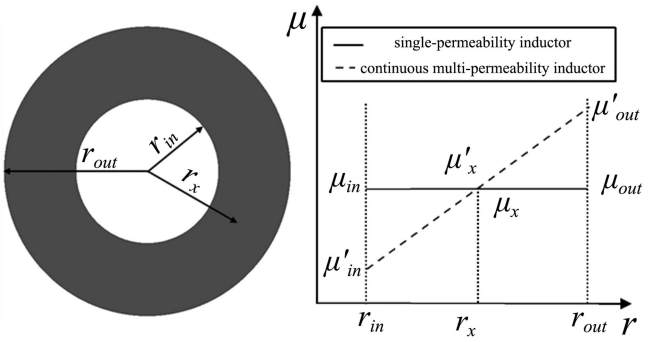


Fig. 3. Schematic diagram of a toroid core and the permeability versus radius of a single-permeability core and a CVMP toroid core.

A. Magnetic Flux Density

Our purpose is to design a CVMP toroid core with the permeability varying along its radius to achieve a constant flux density. Fig. 3 shows schematically the designed permeability versus the radius of the commercial single-permeability toroid core and the CVMP toroid core in order to make the proposed CVMP toroid core capable of being saturated at the same time. The permeability of the commercial single-permeability toroid core and the CVMP toroid core at r_x is expressed as μ_x and μ'_x, respectively. Then, the corresponding *B* of the commercial single-permeability core and the proposed CVMP core at r_x is expressed by B_x and B'_x, respectively.

For the commercial single-permeability toroid core, the permeability is identical anywhere

$$\mu_{\rm in} = \mu_x = \mu_{\rm out}. \tag{3}$$

The magnetic flux density (*B*) at r_x can be expressed by

$$B_x = \frac{\mu_0 \mu_{\rm in} I}{2\pi r_x}. \tag{4}$$

For the CVMP toroid core, the permeability is changeable with the radius

$$\mu'_{\rm in} \neq \mu'_x \neq \mu'_{\rm out}. \tag{5}$$

However, the magnetic flux density (*B'*) in the CVMP core should be identical anywhere in order to be treated as a constant flux inductor. The magnetic flux density can be expressed by

$$B'_{\rm in} = B'_x = B'_{\rm out}. \tag{6}$$

When we substitute (1) into (6), the ratio of the permeability to the radius should be identical

$$\frac{\mu'_{\rm in}}{r_{\rm in}} = \frac{\mu'_x}{r_x} = \frac{\mu'_{\rm out}}{r_{\rm out}}. \tag{7}$$

That is to say

$$\mu'_x = \mu'_{\rm in} \frac{\rm r_x}{r_{\rm in}}. \tag{8}$$

According to (2), it is possible to obtain a CVMP toroid core obeying the relationship of (8) by sintering the proposed core at the specific gradient temperature.

B. Inductance of a Single-Permeability Core and a CVMP Toroid Core

The inductance is vital to an inductor because it represents the energy storage capacity and further affects the power density of the converter [26]. For a toroid core, the per unit thickness inductance can be expressed by

$$dL = N^2 \frac{\mu_0 \mu_r h dr}{2\pi r_x} \tag{9}$$

where *N* is the turns around the toroid core, μ_0 is the permeability of vacuum, μ_r is the relative permeability of the core, *h* is the height of the toroid core, r_x is the corresponding radius of the core, *dr* is the thickness of the concentric ring, and *dL* is the inductance of the concentric ring with the ring thickness of *dr*.

For a single-permeability toroid core, the permeability is identical everywhere, as shown in (3), then the inductance can be expressed by

$$L_{\rm single} = N^2 \frac{\mu_0 \mu_{\rm in} h}{2\pi} \int_{r_{\rm in}}^{r_{\rm out}} \frac{1}{r_x} dr. \tag{10}$$

After integrating, (10) could be also expressed as follows:

$$L_{\rm single} = N^2 \frac{\mu_0 \mu_{\rm in} h}{2\pi} \left(\ln r_{\rm out} - \ln r_{\rm in} \right). \tag{11}$$

For a CVMP constant flux toroid core, the permeability varies with the radius, and the inductance can be expressed by

$$L_{\rm multi} = N^2 \frac{\mu_0 \mu'_x h}{2\pi r_x} \int_{r_{\rm in}}^{r_{\rm out}} dr. \tag{12}$$

Based on (7), we could obtain the expression as follows:

$$L_{\rm multi} = N^2 \frac{\mu_0 \mu'_{\rm in} h}{2\pi r_{\rm in}} \left(r_{\rm out} - r_{\rm in} \right). \tag{13}$$

In order to compare $L_{\rm single}$ with $L_{\rm multi}$, we can obtain the following relationship by dividing (11) with (13):

$$\frac{L_{\rm single}}{L_{\rm multi}} = \frac{\mu_{\rm in} \left(\ln r_{\rm out} - \ln r_{\rm in} \right)}{\frac{\mu'_{\rm in} (r_{\rm out} - r_{\rm in})}{r_{\rm in}}}. \tag{14}$$

Here, assuming $r_{\rm out} = a r_{\rm in}$ ($a > 1$), (14) can be simplified as

$$\frac{L_{\rm single}}{L_{\rm multi}} = \frac{\mu_{\rm in} ln a}{\mu'_{\rm in} (a-1)}. \tag{15}$$

According to the (15), it is possible to realize $L_{\rm single} = L_{\rm multi}$ before the magnetic flux saturation of the core as long as $\mu_{\rm in} \ln a = \mu'_{\rm in}(a-1)$. That is to say, the relationship between $\mu_{\rm in}$ of a single-permeability core and $\mu'_{\rm in}$ of a multi-permeability core could be determined to satisfy $L_{\rm single} = L_{\rm multi}$ as long as the "*a*" (i.e., $r_{\rm out}$ and $r_{\rm in}$) is given according to (15) before the core saturates. It can be speculated that the inductance of a CVMP core has the same inductance with lower current excitation and higher inductance with larger current due to its variable permeability compared with the single-permeability core. The variable permeability could extend the saturation current and decrease heat production, and may further improve the output efficiency of the converter.

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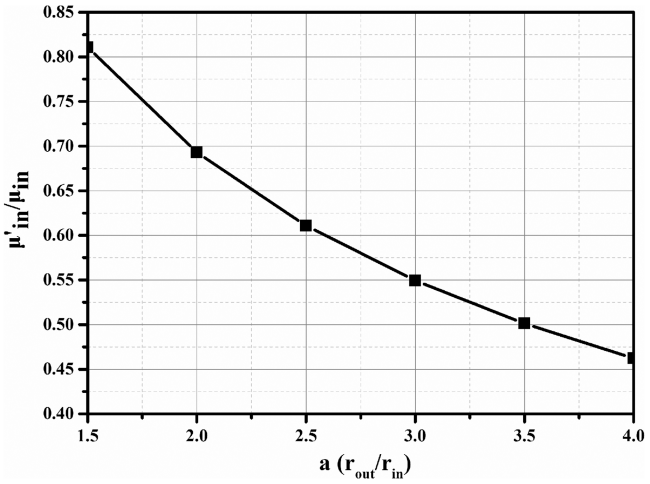


Fig. 4. $\mu'_{\rm in}/\mu_{\rm in}$ as a function of "a" ($r_{\rm out}/r_{\rm in}$).

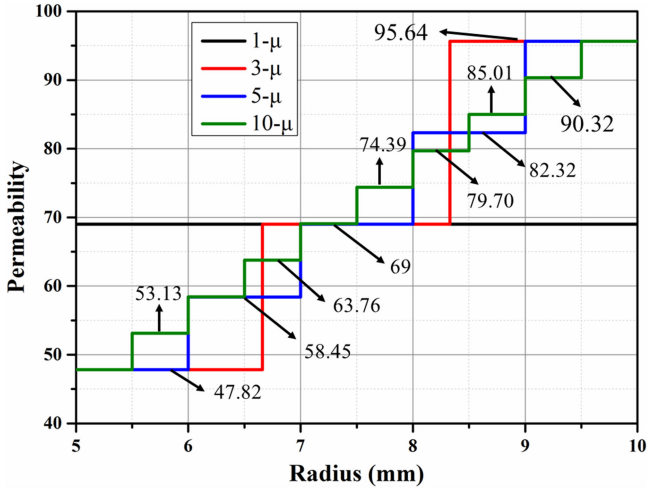


Fig. 5. Designed permeability of 1-μ core and multi-permeability toroid cores.

When $L_{\rm single} = L_{\rm multi}$

$$\frac{\mu_{\rm in} \ln a}{\mu'_{\rm in}\,(a-1)} = 1. \tag{16}$$

For further calculating

$$\frac{\mu'_{\rm in}}{\mu_{\rm in}} = \frac{\ln a}{a-1}. \tag{17}$$

Fig. 4 shows the value of $\mu'_{\rm in}/\mu_{\rm in}$ as a function of "a" under the condition that $L_{\rm single} = L_{\rm multi}$. We can see the ratio of $\mu'_{\rm in}/\mu_{\rm in}$ is approximate linear with the "a" ranging from 1.5 to 4. The scope of the "a" from 1.5 to 4 is enough considering that the ratio of $r_{\rm out}/r_{\rm in}$ of a commercial toroid core is not too large [27], [28]. We can see the value of $\mu'_{\rm in}/\mu_{\rm in}$ is decreased with the "a" increases from Fig. 4. That is to say the permeability ratio of inner side of the CVMP toroid core to the single-permeability core should be lower with the toroid core gets thicker to maintain the same inductance of the two cores.

III. Flux Density Distribution and Inductance Simulation

In this section, the Ansoft Maxwell 16.0 was used to further demonstrate the distribution of magnetic flux density in a single-permeability core and our proposed multi-permeability toroid core. However, we could not build the model with continuously variable permeability versus radius of the core due to the limitation of the Maxwell software. Therefore, the relationship of discretely variable permeability versus radius was established. The models of a multi-permeability toroid core were divided into several (i.e., 3, 5, 10) regions with different permeability from inner side to outer side. The reason that ten-permeability toroid core model established was to simulate the CVMP toroid core as much as possible. All of the model cores have the same size, i.e., $r_{\rm in} = 5$ mm, $r_{\rm out} = 10$ mm, $h = 3$ mm. That is to say, the value of "a" is 2. Hence, the permeability relationship between single-permeability and multi-permeability toroid cores was also determined according to (17). The single-permeability and multi-permeability cores were expressed by 1-μ, 3-μ, 5-μ, and 10-μ, respectively. The permeability of the 1-μ core was set to be 69 according to the NiZnCu ferrite in [25]. The designed permeability of the multi-permeability toroid cores was from 47.82 to 95.64 based on (17), as shown in Fig. 5. Meanwhile, the inductance versus dc was also analyzed.

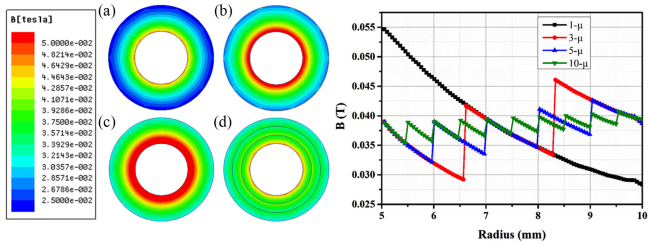


Fig. 6. Simulated magnetic flux density (B) distributions and B versus r of (a) 1-μ core, (b) 3-μ core, (c) 5-μ core, and (d) 10-μ core.

A. Magnetic Flux Density Distribution

The following example compares the simulated flux density distribution of a single-permeability toroid core with that of the multi-permeability toroid cores. The 1-μ core, 3-μ core, 5-μ core, and 10-μ core are indicated in Fig. 6(a), (b), (c), and (d), respectively. For the 1-μ core, the inner side of the core reached as high as 0.055 T, and the outer side could be as low as 0.028 T. It can be seen that the flux density distribution in 1-μ core is not uniform (about 49.1% difference). The results show that the inner side of 1-μ core could be easily saturate, while the flux density of outer side is still very low. For the 3-μ core, the flux density of inner side is not too high, and the outer side is not too low. It shows narrower difference between inner side and outer side compared with that of the 1-μ core due to its incremental permeability versus the radius. For the 5-μ core, the phenomenon becomes more obvious owing to the more layers and smaller difference of the permeability of each layer compared with the 3-μ core. Until to 10-μ core, it presents more uniform flux distribution from the inner wall to outer wall due to the smaller and

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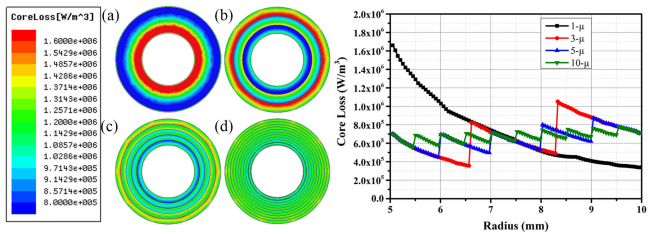


Fig. 7. Simulated core loss density distributions and core loss density versus r of (a) 1-μ core, (b) 3-μ core, (c) 5-μ core, and (d) 10-μ core.

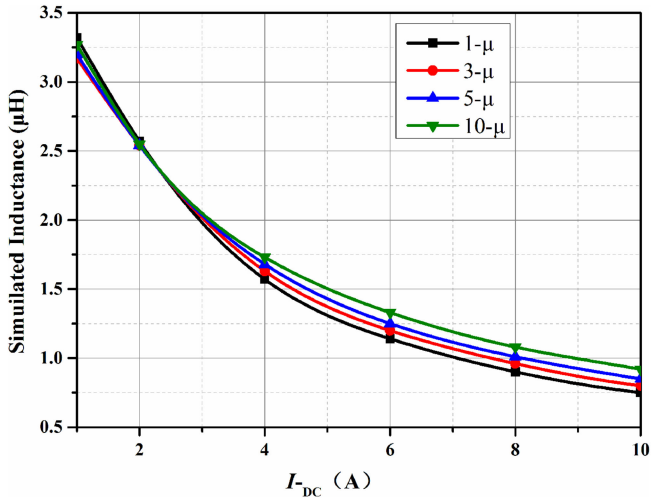


Fig. 8. Simulated inductance at room temperature of the single-permeability and multi-permeability toroid cores versus dc.

smaller difference of the permeability of each layer. It is helpful to maintain the same flux density of each layer and reach to saturation simultaneously from inner side to outer side of the core. It can not only prevent the premature saturation of the inner core but also make full use of the materials of the outer core by this way. For the uniformity of flux density distribution, we can see the core of 10-μ > 5-μ > 3-μ > 1-μ. The more layer, the more uniform. Therefore, it can be speculated that the flux density distribution of a CVMP toroid core will be more uniform than that of a discretely variable permeability toroid core.

B. Core Loss Density Distributions

The simulated core loss density of the single-permeability and the three multi-permeability cores is shown in Fig. 7. The excitation current was in the form of sinusoidal waves with 1-MHz frequency. The electrical resistivity of the core was set as $3.06 * 10^5$ Ω·m. The skin depth varies from 28.73 to 40.63 m based on equation of skin depth, which is much larger than the dimensions of the core ($r_{\text{in}} = 5$ mm, $r_{\text{out}} = 10$ mm, $h = 3$ mm). Therefore, the skin depth could be ignored here. For the 1-μ core, the inner side of the core reached as high as $1.66 * 10^6$ W/m^3, and the outer side could be as low as $3.39 * 10^5$ W/m^3. It can be seen that the core loss density distribution in 1-μ core is not uniform (about 79.57% difference). For the 10-μ core, the core loss density at the inner wall is almost the same as that at the outer wall. It is reasonable that the core loss density could be uniform as a constant if the core could be with CVMP.

C. Inductance Versus DC

Fig. 8 shows the simulated inductance versus dc of a single-permeability and the multi-permeability toroid cores. The results show that the 1-μ core and other three cores have the similar inductance due to their similar average permeability when the dc is less than 2 A. However, the inductance of 1-μ core drops faster than that of other three cores because its single higher permeability saturates earliest with increasing excited dc. That is to say, the saturation problem occurs instead of reducing the permeability and further results in the lower inductance at larger dc. The multi-permeability cores show the advantage when the dc exceeds 2.5 A. The advantage becomes more obvious with the increase of the layers. It can be attributed to the variable permeability from inner side to outer side of the core. In multi-permeability cores, the permeability of outer side is higher which could be helpful for increasing the inductance at lower dc. With increasing the dc, the inner side of the core is not easy to be saturated due to its lower permeability. That is to say, it is the permeability of inner side which plays an important role in keeping the inductance versus dc. In whole load range, the simulated results show that the n-μ core has a capacity to take a higher dc compared with the 1-μ core. And the ability of taking high dc gets bigger as the n grows.

IV. Prototype Fabrication and Experiment

To identify the performance in practice, the CVMP toroid core was fabricated. We introduced how to demonstrate the proposed CVMP core in this section. For the purpose of comparison, a single-permeability toroid core was also fabricated as a benchmark. The two kinds of cores have the same dimension, i.e., 20 mm in outer diameter, 10 mm in inner diameter, and 3 mm thick ($r_{\text{out}} = 10$ mm, $r_{\text{in}} = 5$ mm, $h = 3$ mm).

A. Principle on How to Fabricate a CVMP Toroid Core

It has been proved that the permeability of NiZnCu ferrite (Product ID: LSF50, Powder Processing & Technology, LLC, Valparaiso, Indiana, USA) was dependent on the sintering temperature as reported in our previous study [25]. The results show that the permeability almost increased linearly with the sintering temperature, as shown in Fig. 2. Therefore, it is possible to fabricate a CVMP toroid core using only one magnetic material by controlling the distribution of sintering temperatures precisely from inner side to outer side of the core.

The permeability of the single-permeability toroid core was set to be 69. Therefore, the designed permeability of the CVMP toroid core was set to be from 47 to 94 from inner side to outer side of the core based on (8) and (17). Meanwhile, the theoretical temperature gradient was determined to be from 890 to 1020 °C from inner side to outer side of the core according to (2). Before fabricating the cores in practice, we have simulated the temperature field according to the presupposed condition

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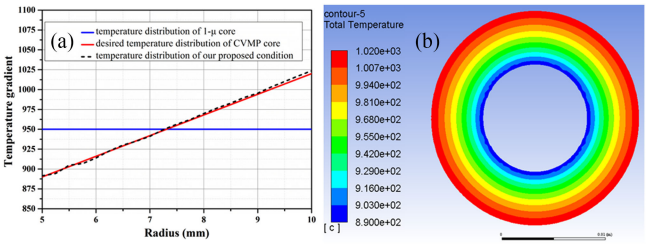


Fig. 9. (a) Desired and simulated temperature gradient versus the radius of the core. (b) Distribution of simulated temperature gradient of the CVMP core according to our proposed condition.

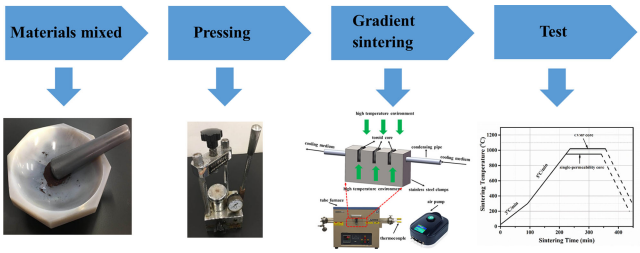


Fig. 10. Schematic diagram of fabricating the CVMP toroid core.

using the Fluent software. Owing to the principle of radiation heating of the tube furnace, the restrained boundary conditions was equivalent to the tube furnace. The cooling air with a flow rate of 95 L/min was set to be flowed from one side and come out from the other side of the pipe. We can see the simulated temperature gradient and theoretical temperature gradient are almost coincided from Fig. 9(a). The deviations are mainly near the core center and outside of the core; it is mainly due to the convective coefficient and absorbing coefficient between air and the core materials. The little deviations will not affect the permeability obviously and can be ignored. Fig. 9(b) shows the distribution of the simulated temperature gradient. In conclusion, the desired temperature gradient could be realized in theory as long as the air rate was moderate. It can be speculated that the design method and manufacturing condition is reasonable.

Fig. 10 shows the schematic diagram of the whole fabrication process to make the CVMP cores. The LSF50 powders were mixed uniformly with the PVA binder and the mixture was pressed into a green body with toroidal shape using an axial tablet press. Three prepared green bodies of the toroid core were placed into the designed groove of the stainless steel clamps at one time, as shown in Fig. 10. The stainless steel cooling pipe with the outer diameter of 7 mm and the inner diameter of 5 mm was used to flow the coolant, e.g., air or water, by a pump. The cooling pipe was run through the stainless steel clamps and cross the center of the green bodies which were placed on the stainless steel clamps. During the sintering, the furnace temperature could be set as the maximum sintering temperature and the sintering temperature of the inner wall of the CVMP cores could be adjusted by the flowing rate of the coolant. Then, the temperature gradient along the radius of the cores could be generated as desired. In other words, we could obtain the desired

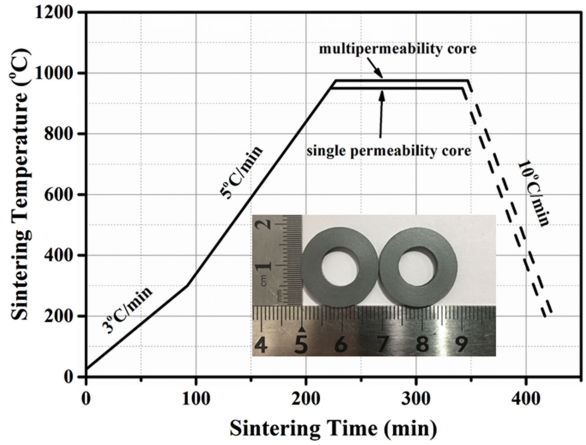


Fig. 11. Sintering temperature curve and prototype of the sintered cores.

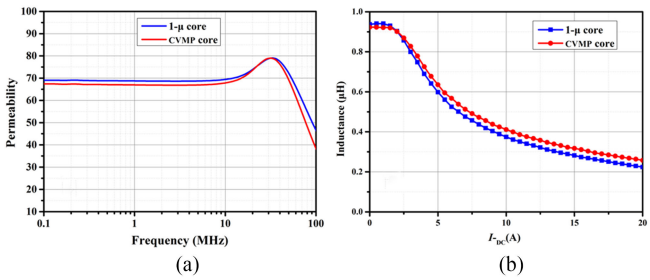


Fig. 12. (a) Permeability of single-permeability and CVMP toroid cores versus frequency. (b) Inductance of single-permeability and CVMP toroid core versus dc bias at 1 MHz.

permeability gradient in the cores by generating the specific temperature gradient.

B. Fabrication of CVMP Toroid Core

We fabricated a CVMP core with the temperature gradient from 890 to 1020 °C from inner side to outer side of the core to compare with that of a single-permeability core sintered at 950 °C for a fair comparison. The sintering curve and the prototypes of the single-permeability core and the CVMP core are shown in Fig. 11. In this experience, the inner side and outer side temperature of CVMP core was measured by two thermocouples. The test results showed that the temperature gradient from 890 to 1020 °C from inner side to outer side of the core could be realized with the air flow rate of 95 L/min. It is worth noting that the single-permeability core does not need the device to assist sintering.

C. Experimental Results

1) Permeability and Inductance Versus DC Bias: The permeability (μ_r) of the cores was characterized by an impedance analyzer (E4990A, Agilent) with a test fixture (16454A, Agilent) in a frequency ranging from 100 kHz to 100 MHz. The results are shown in Fig. 12(a). μ_r of the single-permeability core sintered at 950 °C and the CVMP core is 69 and 67, respectively. It should be an error which is caused by the actual sintering

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process. It can be tolerated and ignored. The two sintered cores were wound using five turns copper wires for inductance measurement. And the inductance versus dc bias was also tested by a combination of the impedance analyzer and a dc-bias fixture. The test frequency is 1 MHz.

It can be seen that the inductance before 2 A of the single-permeability core and the CVMP core is 0.93 and 0.91 μH, respectively, as shown in Fig. 12(b). It is in accordance with (9) which is due to the μ_r of the single-permeability core is a little higher than that of the CVMP core shown in Fig. 12(a). While the CVMP core shows the advantages with the dc increasing, it is mainly because the single-permeability core has higher permeability with the lower current exciting and reaches saturation earlier than the CVMP core. The permeability will decrease with the increase in dc, which could further decrease the inductance. The higher inductance of the CVMP core at larger dc-bias current is attributed to the continuously variable permeability along the radius of the CVMP core. It was concluded that the way to design the CVMP toroid core could be able to increase the operating current than the single-permeability core. The tested results are consistent with the simulated results.

2) Core Loss Density: The core loss density presents the energy conversion efficiency and the heat production of an inductor in a converter [29]. It is important to reduce the core loss density of a toroid core for designing a high-power-density converter [30], [31]. The core loss calculation model is usually in the form as follows [32], [33]:

$$P_v = C_m f^{\alpha} B_m^{\beta} \tag{18}$$

where P_v is the power loss per unit volume, f is the frequency, B_m is the ac flux density magnitude, and C_m, α, β are Steinmetz parameters that depend on materials. However, the inductors are operated in a converter with dc bias in most cases. The conventional Steinmetz (18) is no longer applicable due to the fact that dc bias has a strong impact on the value of core loss density. Therefore, the core loss density was further tested.

The core loss density was measured by a self-built system. The principle of the core loss density measurement had been reported in [34] and [35]. The principle of the setup is to cancel the reactive power in the core under test by using an additional transformer with a lossless or low-loss core to cancel the reactive voltage and reduce the sensitivity to phase discrepancy. Then, the phase discrepancy-induced error could be reduced greatly, and accurate measurement could be achieved for the higher operation frequency in this way rather than the classical four-wire method. Furthermore, the measurement circuit is wideband, so it can measure the loss of arbitrary waveform like triangular flux, which is the practical interest of power electronics applications [34], [35]. The core loss density of each of the wire-wound cores, i.e., the single-permeability toroid core and the CVMP toroid core, was measured in the form of sinusoidal waves with 1 MHz. Fig. 13 shows the core loss density of both the single-permeability toroid core and the CVMP toroid core ranging from 1 to 10 mT. The results show that the CVMP toroid core has about 20%–30% lower core loss density than that of the single-permeability core. This is mainly attributed to variable permeability of the CVMP toroid core. It could prevent the premature saturation of the inner side of the CVMP core at larger dc bias. As a result, it is beneficial to reduce the core loss density. The lower core loss density means lower energy waste in power conversion.

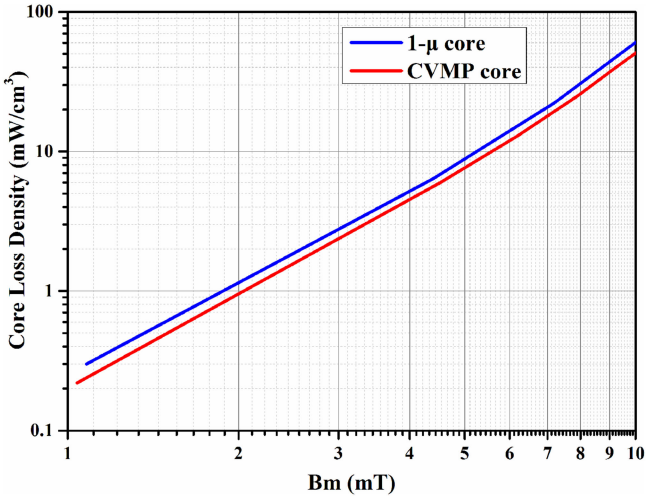


Fig. 13. Core loss density of single-permeability and CVMP toroid cores measured at 1 MHz.

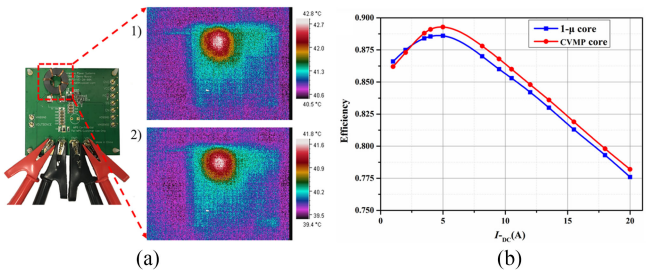


Fig. 14. Thermal picture and the efficiency curves of the dc–dc converter with the two cores.

3) Efficiency in a DC–DC Converter: A 1-MHz 8-V input, 1-V output dc–dc converter was constructed to test the two-core prototypes. Fig. 14(a) and (b) shows the thermal pictures of the two cores and the output efficiency of the converter, respectively. The output efficiency of the converter could be obtained through dividing the product of the output voltage and output current by the product of the input voltage and input current. In Fig. 14(a), the single-permeability and the CVMP cores were shown as a-1 and a-2, respectively. Both of them were tested under the same condition which operated at 5 A and 5 min. It can be seen that the temperature of the single-permeability core was 42.8 °C, which was 1 °C higher than that of the CVMP core. That means the single-permeability core produced more heat and wasted more energy than the CVMP core.

The single-permeability core and the CVMP core were applied in the 1-MHz 8-V input, 1-V output dc–dc converter separately. A voltage source with an 8 V output was connected with the input terminal of the dc–dc converter. Then, the dc–dc converter could output 1 V stably. Different resistors were used to control the output current. The output efficiency of the converter

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with the different cores could be obtained through dividing the product of the output voltage and output current by the product of the input voltage and input current. From Fig. 14(b), we can see the efficiency of the converter with the CVMP toroid core was higher than that of single-permeability core at larger than 2.5 A, which means that the CVMP toroid core can help to improve the efficiency of the converter at heavy load. Fig. 14 shows that the output efficiency of the CVMP core, i.e., $\sim$89%, is $>$1% higher than that of the single-permeability core when the current exceeds 4 A, benefiting from the lower loss density of the CVMP core. The improvement of the output efficiency at 1 MHz should be attributed to the lower loss density of the CVMP core because only the single-permeability core was replaced by the CVMP core in the dc–dc converter. And the inductance was identical between the CVMP and the single-permeability one. The 1% improvement could be critical to demonstrate the feasibility of the proposed way to make a CVMP core, especially for the large current excitation. Similar methods on the efficiency measurement of an inductor were reported in [18] and [19].

V. Conclusion

This paper proposed a CVMP toroid core to improve the uniform distribution of magnetic flux density by precision controlling the sintering temperature from the inner side to outer side of the core. The results show that the CVMP toroid core has lower core loss density without sacrificing the inductance than a single-permeability core. The output efficiency of the CVMP toroid core employed in a dc–dc converter could be improved due to the variable permeability from inner side to outer side of the core with same volume at heavy load. The variable permeability means that it can avoid magnetic saturation earlier and reduce the heat production near the core center and fully used the materials outside of the core. That is to say, the CVMP toroid core has the capacity of taking higher current than a single-permeability core.

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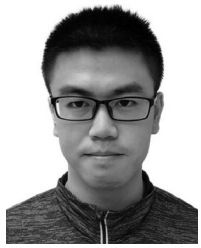
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